AGST 50104. Experimental Design

Instructor:
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Location and Time:
Lecture: Monday & Wednesday 10:45 AM – 12:00 PM (PTSC 007)
Lab: Friday 10:45 AM – 12:35 PM (AFLS B108)

Office Hours:
Monday & Wednesday, 1:00 PM – 2:00 PM (PTSC 107) or by appointment

Course Information

Course Objectives

By the end of this course, students will be able to:

- Apply the fundamental principles of experimental design (replication, randomization, and blocking) to plan efficient and valid experiments in agricultural and biological research
- Identify appropriate experimental designs (CRD, RCBD, factorial, split-plot, incomplete block) based on experimental constraints, research questions, and sources of variability
- Formulate research hypotheses as testable statistical models with proper specification of fixed and random effects
- Conduct rigorous statistical analyses using ANOVA, linear mixed models, and generalized linear models with appropriate diagnostics and assumption checking
- Interpret statistical output critically, distinguishing between statistical significance and practical importance, and communicate results effectively to scientific audiences

- Recognize common pitfalls in experimental design and analysis (pseudoreplication, confounding, inappropriate error terms) and propose solutions

Lab Objectives

Through hands-on practice in lab sessions, students will:

- Develop basic proficiency in R programming using tidyverse workflows for data manipulation, visualization, and statistical analysis
- Generate randomized experimental designs using R packages and document design decisions
- Implement complete analysis workflows: exploratory data analysis → model fitting → diagnostics → inference → visualization of results
- Create reproducible reports and presentations using Quarto that integrate code, analysis, and interpretation
- Critique and troubleshoot analyses from real agricultural/biological experiments, identifying design flaws and suggesting improvements

Computing

I will primarily use R output in lectures, homework, and exams. R scripts will be provided on class materials. Statistical software, such as JMP, SPSS, and MATLAB, can be used for homework and the project, but I will not be responsible for providing technical support for those.

Prerequisites

AGST 50203 or STAT 51303 (or equivalent).

Course Structure

- 30% (300 points) Weekly Homework
- 15% (150 points) Exam I (end of Part I)
- 15% (150 points) Exam II (end of Part II)
- 20% (200 points) Final Exam (comprehensive)
- 20% (200 points) Final Project

*Please see the schedule below.

Grading Scale:

The table below shows the approximate distribution of letter grades one should expect at the end of the semester. Total of 1,000 points will be distributed across quizzes, homework, and exams. Your total points at the end of the semester will be rounded to the nearest tenth of a point. For example, a grade of 899.4 will be rounded to 899, and that is a B. Conversely, a grade of 899.5 will be rounded to 900, and that would be an A. No rounding will be done before the final grade determination. At the end of the semester, the highest grade will be adjusted to 1,000 and all others will increase proportionally.

Letter Grade	Point Range	Percentage
A	900–1000	90–100%
B	800–899	80–89%
C	700–799	70–79%
D	600–699	60–69%
F	0–599	0–59%

Homework:

Team Structure:

Homework will be completed in **groups of ~5 students** formed in Week 1. Groups work together on all homework assignments and the final project throughout the semester. The instructor will create a Microsoft Teams chat for each group to facilitate communication and provide support.

Assignment Frequency:

New homework assignments are posted (almost) every Friday via Teams. Follow in-class instructions for any exceptions.

Two-Part Submission:

1. *Written Submission (Due Thursday 11:59 PM)*

- All group members should submit the same file to Blackboard
- Quarto (you will learn this on Week 1) document (`.qmd`) + rendered PDF
- File naming: `group#_hw#.qmd` and `group#_hw#.pdf` (e.g., `group1_hw3.pdf`)
- Peer evaluation for all group members (see details below)

2. *Oral Presentation (Friday Lab)*

   - One group member is randomly selected to present
   - Presenter explains one question (selected by the instructor) from the submitted homework
   - Time limit: 5 minutes maximum
   - Instructor projects the submitted PDF during presentation

Grading Breakdown:

Your homework grade consists of **group** and **individual** components:

Component	Weight	Grading Basis
Written Submission (Group)	60%	Quality, correctness, completeness, clarity of analysis and visualizations
Oral Presentation (Individual)	20%	Presenter's clarity, correctness, and ability to explain the group's work
Participation & Contribution (Individual)	20%	Peer evaluation (group members)

Grade Distribution:
- **Group component (60%)**: All group members receive the same score based on the written submission
- **Individual presentation (20%)**: Only the randomly selected presenter is graded; this score applies to that individual's homework grade. For each student, this portion of the grade will be the average of all times you presented, i.e., if you presented 2 times and got 60/100 and 40/100, at the end of the semester you will get 50/100 for the individual portion. If you only presented once and got 70/100, that will be your grade for the individual presentation component.
- **Individual participation (20%)**: Based on peer evaluations. Each student should submit to

Board along with the PDF document, a CSV file rating (from 1-10 based on contribution to group work, communication, reliability, quality of contributions) each group member in terms of contribution to the homework conclusion. At the end of the semester, the instructor will use the average rating provided by each group member to calculate the score on this component of the homework.

Presentation Policy:
- If you are **absent when selected**, you automatically become the presenter for the next homework (no random selection).
- If a given student is not randomly selected after a few rounds, the instructor will pick that student to present, so each student has the chance to present.

Policies:
- *No late submissions*: Submit partial work rather than nothing
- *No make-ups*: Exceptions only if discussed one week in advance
- *Lowest homework dropped*: Your lowest homework score is excluded from the final average (allows one missed homework)
- *Solutions*: Discussed during Friday lab after presentations

Exams:

Exams will include topics covered until that time. Although exams 1 and 2 will not necessarily be cumulative, concepts covered in initial classes are essential to topics covered late in the semester. The final exam can include any topic covered in class. The final exam will take place on **Wednesday, May 6, 2026 10:15 AM to 12:15 PM**. All other exams will take place during lab time. Exams 1 and 2 will be worth 150 points, whereas the final exam will be worth 200 points. Exams are closed book; however, you will be allowed to bring your own formula's sheet on a SINGLE paper (only 1 side and written with your OWN HAND, i.e., no printed pages). The exam could be hand-written or coded, you should be prepared for both. Each exam focuses on the course material covered since the previous exam but it will still contain earlier material due to the repetition of concepts. If you cannot attend an exam at the assigned time because of an unplanned but legitimate reason (e.g., sickness, family emergency), a make-up exam will be scheduled. If you have another reason, such as a job interview or plan to attend a conference, you must notify me at least two weeks prior to the exam in order to decide on the legitimacy of the reason and the possibility of rescheduling the exam. Arranging cheap airfare home prior to the final is never a legitimate reason.

Group Project:

The teams formed in Week 1 will complete an independent project focused on one of the experimental design topics covered in class. The topic will be randomly assigned in Week 1.

Format: Groups will record a 9-11 minute video presentation and post it on YouTube (unlisted). The link should be submitted to the instructor anytime during the semester, but no later than **April 24, 2026, at 11:59 PM**.

Grading: A comprehensive rubric with specific criteria and point allocations is provided in the separate document: `group_project_rubric.pdf` (available on Blackboard and the course materials repository).

Important: Only group members who appear in the video will receive a grade for the project.

AI Guidelines:

Students are encouraged to use AI tools, such as GitHub Copilot, to assist with studying, completing homework, and understanding course material. These tools can be valuable for learning programming techniques, exploring statistical concepts, and improving reproducibility in analysis workflows. However, YOU are responsible for any content you submitted, not the AI you used. Unless explicitly authorized, **AI CANNOT BE USED** on exams and any unauthorized use will be considered a violation of the University's Academic Integrity Policy. Students must complete exams independently, without external assistance, to ensure a fair assessment of their knowledge and skills.

Course Requirements:

You will need a computer for all homework assignments. While the instructor can give you guidelines, it is YOUR RESPONSIBILITY to make sure that you have access to R and can render Quarto documents (Posit Cloud is a free resource the instructor will show). Your homework must have the exercises presented in order. These solutions should include all relevant graphs and tables appropriately labeled and described. Providing extra output and not highlighting the important information or providing hard to follow answers can result in a loss of points.

Time Management:

Procrastination is often fatal to statistical homework. Students will benefit from making progress on assignments before the last class of the week, so they may ask questions and work on assignments during lab sessions.

Text book

- **Primary text:** Oehlert, G. W. (2010). A First Course in Design and Analysis of Experiments. (Freely available on the author's website: http://users.stat.umn.edu/~gary/Book.html
- **Complementary resources:**
    - Lawson, R. (2014). Design and Analysis of Experiments with R. CRC Press. ISBN 9781439868133
    - Quinn, G. P. & Keough, M. J. (2002). Experimental Design and Data Analysis for Biologists. Cambridge University Press.
- **Other useful references:**
    - Biostatistical Design and Analysis Using R: A Practical Guide. Logan, 2010. ISBN:9781444319620 | DOI:10.1002/9781444319620 (Available online through the U of A Library)

Software

- **R:** https://cran.r-project.org/
- **VS Code:** https://code.visualstudio.com/
- **Quarto:** https://quarto.org/
- **Posit Cloud:** https://posit.cloud/ *(free online alternative if local setup is problematic)*
- **RStudio:** https://rstudio.com/products/rstudio/download/

Tentative Schedule

Week 1 (1/12, 1/14, 1/16)

- Course overview and Introduction to Experimental Design
- VS Code setup and configuration for data science
- R programming with tidyverse for data manipulation and visualization
- Quarto for reproducible reports and presentations
- GitHub account setup and GitHub Copilot for AI-assisted programming

Week 2 (1/21, 1/23) *(Mon 1/19 MLK Day — no class)*

- Completely Randomized Design (CRD) — ANOVA basics
- Assumptions and diagnostics
- Contrasts and multiple comparisons

Week 3 (1/26, 1/28, 1/30)

- CRD continuation: orthogonal contrasts and power considerations
- Randomized Complete Block Design (RCBD) — Part 1
- Blocking principles and layout

Week 4 (2/2, 2/4, 2/6)

- RCBD — Part 2: handling missing plots, additivity (Tukey 1-df)
- Agricultural examples and diagnostics

Week 5 (2/9, 2/11, 2/13)

- Factorial designs — Part 1: 2x2 and basic 2^k designs
- Main effects, interactions, and interaction plots

Week 6 (2/16, 2/18, 2/20)

- Factorial review and design choices
- Review session (covering Weeks 1–5)
- **Exam I** — Friday, February 20, 2026 (covers Weeks 1–5)

Week 7 (2/23, 2/25, 2/27)

- Split-plot and strip-plot designs — Part 1
- Whole-plot vs subplot factors; randomization and EMS intuition

Week 8 (3/2, 3/4, 3/6)

- Split-plot and strip-plot designs — Part 2
- Mixed-model formulation for split-plot; variance components

Week 9 (3/9, 3/11, 3/13)

- Linear Mixed Models — Part 1
- Random vs fixed effects, REML vs ML, ICC

- Random intercept models and diagnostics

Week 10 (3/16, 3/18, 3/20)

- Linear Mixed Models — Part 2
- Random slopes, crossed vs nested effects
- Repeated-measures covariance structures (CS, AR1)

🌴 Spring Break (March 23–27, 2026) — No Classes

Week 11 (3/30, 4/1, 4/3)

- Incomplete block designs (BIBD and alpha designs)
- Design efficiency and analysis; mixed-model framing
- Review Session — Wednesday, April 1, 2026
- **Exam II** — Friday, April 3, 2026 (covers Weeks 6–11)

Week 12 (4/6, 4/8, 4/10)

- Repeated measures and longitudinal applications
- Handling missingness in longitudinal data; subject-level diagnostics

Week 13 (4/13, 4/15, 4/17)

- ANCOVA: covariates in designed experiments
- When covariates help or hurt; interpretation and diagnostics

Week 14 (4/20, 4/22, 4/24)

- Response Surface Methodology (1 week)
- Central Composite and Box–Behnken designs
- Curvature detection, ridge analysis, and contour visualization

Week 15 (4/27, 4/29, 5/1)

- GLM basics: logistic and Poisson regression; offsets; overdispersion

- GLMM introduction: random intercepts in GLM; convergence cautions
- Integration, reporting, and reproducible Quarto workflows (course synthesis)

Final Exam — Wednesday, May 6, 2026 10:15 AM to 12:15 PM

Course Policies

Recording of Class Lectures:

By attending this class, students understand the course is being recorded and consent to be recorded for official university educational purposes. Be aware that incidental recording may also occur before and after official class times. Recordings will be available on Echo360. The link is available on Blackboard.

Unexcused Absences and Make-Up Work:

Absences will be considered excused in cases of inclement weather (see below), death in the family, illness in the immediate family, personal illness, certain University-sponsored activities, religious observances, jury duty (or subpoena for court appearance), or military duty. In general, the claim of personal illness must be accompanied by a letter from the student's physician describing the nature of the illness. Absence from class for any reasons other than those listed above will be considered unexcused.

Academic Integrity:

As a core part of its mission, the University of Arkansas provides students with the opportunity to further their educational goals through programs of study and research in an environment that promotes freedom of inquiry and academic responsibility. Accomplishing this mission is only possible when intellectual honesty and individual integrity prevail. Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/. Students with questions about how these policies apply to a particular course or assignment should immediately contact their instructor.

Student Conduct:

The University of Arkansas, Student Standards of Conduct, as outlined in the Student Handbook (http://handbook.uark.edu/standardsofconduct.php) will be complied with in this class. The class atmosphere must be conducive to collaborative learning. Class disruptions, including, but not limited to, excessive absences, tardiness, failure to assume responsibility for cooperative learning, or disturbances in class, will be penalized by a reduction of class participation points and/or dismissal from class.

ADA Accommodations:

If you need accommodation due to a disability, please schedule a meeting to discuss this with the instructor sometime during the first two weeks of class or as soon as possible upon diagnosis. Students must be registered with the Center for Educational Access (http://www.uark.edu/ua/csd/; 575-3104; 575-3646; 104 Arkansas Union; ada@uark.edu) and hand-deliver an official Accommodation Letter from the Center for Educational Access for accommodations.

Inclement Weather Policy:

In the event of inclement weather, the class will be canceled if: 1) the University is closed; or 2) the bus service to and around campus has been discontinued. Otherwise, the class will be held. If the University remains open, but road conditions are such that, in your judgment, you cannot safely make it to class, call the instructor as soon as possible, or leave a message with the departmental secretary if the instructor is not immediately available, so that arrangements can be made for you to make up the missed work.

Emergency Procedures:

Many types of emergencies can occur on campus; instructions for specific emergencies such as severe weather, active shooter, or fire can be found at https://safety.uark.edu/emergency-preparedness/.

Contingency Plans:

In the event of campus closure, we will continue utilizing our schedule using Echo360/Blackboard/BOX share drive as the portal for the delivery of course materials and UARK email for communications. Please check both of these areas immediately for guidance and directions from me.

Severe Weather (Tornado Warning):

- Follow the directions of the instructor or emergency personnel.
- Seek shelter in the basement or interior room or hallway on the lowest floor, putting as many walls as possible between you and the outside
- If you are in a multi-story building and you cannot get to the lowest floor, pick a hallway in the center of the building

- Stay in the center of the room, away from exterior walls, windows, and doors

Violence / Active Shooter (CADD):

- **CALL** - 9-1-1
- **AVOID** - If possible, self-evacuate to a safe area outside the building. Follow the directions of police officers.
- **DENY** - Barricade the door with desk, chairs, bookcases, or any items. Move to a place inside the room where you are not visible. Turn off the lights and remain quiet. Remain there until told by police it's safe.
- **DEFEND** - Use chairs, desks, cell phones or whatever is immediately available to distract and/or defend yourself and others from attack.

Blackboard Ultra Information

All course materials, including lecture slides, lab materials, and homework assignments, will be available on Blackboard Ultra. Please log in at https://learn.uark.edu. Ensure you have enabled notifications on Blackboard Ultra to receive updates about assignments, grades, and announcements.

Support Resources

For help with Blackboard Ultra, visit the Blackboard Help Center.